

1 Supporting Information for

2 Multi-omics profiling identified two epithelioid sarcoma molecular subtypes with
3 distinct signaling and immune characteristics

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Supplementary Materials and Methods

Patient samples

Sample selection

Cases with pathological diagnosis of epithelioid sarcoma (EpS) or extracranial extrarenal rhabdoid tumor (EERT) (age >18 years) between January 1998 and January 2022 were retrieved from the Department of Pathology of Institut Gustave Roussy, Villejuif, France. All cases were histologically reviewed by two soft tissue pathologists (C.N. and J-M. C.) and classified by histological subtype (distal, proximal or hybrid EpS) according to the reference consensus guidelines (**Supplementary Table S1**) [1]. Final pathological diagnoses and clinico-pathological characteristics of the patients are summarized in **Supplementary Table S2**. Archived frozen and formalin-fixed paraffin-embedded (FFPE) samples were used according to the French legal framework of MR-004 reference methodology set up by the French Data Protection Authority (Number I14112002202020 Health Data Hub, available at <https://www.health-data-hub.fr/>). Tumor samples were collected regardless of tumor status (untreated or treated primary tumors, recurrences, or metastases), with a preference given to untreated primary tumors whenever available.

An additional external cohort of 12 EpS and 1,041 selected soft tissue tumors' bulk RNA-sequencing (RNA-seq) data was provided by Centre Léon Bérard, Lyon, France.

For single-cell RNA-sequencing analysis

Seven patients among the selected cases signed a preoperative written informed consent to allow the use of tumor residual samples for single-cell RNA-sequencing (scRNA-seq) analysis.

Bulk RNA sequencing

Sample preparation and paired-end sequencing

Total RNAs were extracted from FFPE tissue sections with tumor purity > 40%, using a FormaPure RNA kit (Beckman Coulter, Brea, CA, USA). The Ambion DNase I (Life Technologies, Carlsbad, CA, USA) was used to remove DNA. RNA quantification was assessed using a NanoDrop spectrophotometer (Thermo Fisher

Scientific, Waltham, Massachusetts, USA) measurement and RNA quality using TapeStation system with Hs RNA screen tape (Agilent, Santa Clara, CA, USA). Samples with sufficient RNA quantity ($> 0.5 \mu\text{g}$) and quality [percentage of RNA fragments > 200 nucleotides (DV200) $> 30\%$] were considered suitable for sequencing. Individual libraries were prepared using 100 ng of total RNA and TruSeq RNA Exome kits (Illumina, San Diego, CA, USA). Sequencing was performed (paired-end, 2×75 cycles) using NextSeq 500/550 high output kits on a NextSeq 500 machine, or the NovaSeq 6000 SP/S1 reagent kit on a NovaSeq 6000 machine (Illumina). The mean number of reads per sample was approximately 80 million.

Bioinformatics and data analysis

Alignments were performed using Spliced Transcript Alignment to a Reference (STAR) version 2.6.1 (available at <https://github.com/alexdobin/STAR>) on the hg19 version of the human reference genome. Numbers of duplicate reads were assessed using Picard version 2.23.9 (available at <https://broadinstitute.github.io/picard/>). Samples with a number of unique reads less than 8 million were discarded from further analyses. Fusion transcripts were called by STAR-Fusion version 1.10.1 and Arriba version 2.4.0 (available at <https://github.com/suhrig/arriba>). Expression values were extracted using the Salmon v1.4.0 tool with University of California, Santa Cruz Genome Reference Consortium Human Build 37 (GRCh37) reference. All samples were normalized together using a variance stabilizing transformation with the DESeq2 R package version 1.34.0 (available at <https://bioconductor.org/packages/release/bioc/html/DESeq2.html>).

Consensus hierarchical clustering using the Ward.D2 was performed on the transcriptomic data of 33 EpS and 3 EERT samples using the top 25% most variably expressed genes.

Differential gene expression analysis was performed with the DESeq2 R package. The significance was given by adjusted P -value (P_{adj}) < 0.05 and $\log_2$ fold change ($\log_2\text{FC}$) > 1 . Functional enrichment analysis was performed using ToppFun from the ToppGene suite [2]. Enriched gene sets from Gene Ontology Biological Process (GO BP) with $P_{adj} < 0.05$ were selected.

Gene Set Enrichment Analysis (GSEA) was performed using the normalized expression by median of ratios from DESeq2 with the hallmark gene set collection

from the Molecular Signature Database (MSigDB) (available at <https://www.gsea-msigdb.org/gsea/msigdb/>).

To estimate the abundance of immune cells, we used immunedeconv R package version 2.0.4 to deconvolute immune cell gene expression signatures with Microenvironment Cell Populations (MCP) counter and CIBERSORT on transcripts per million (TPM) values.

Whole-exome sequencing (WES)

Sample preparation and sequencing

WES was performed on 17 tumor samples and 16 tumor/normal sample pairs with tumor purity > 40%.

For 14 tumor/normal pairs and 2 tumor samples, sequence capture and exome sequencing were performed by the Sequencing Platform at Institut Curie, Unit of Genetics (Paris, France). Briefly, genomic DNA was isolated from fresh frozen tissue by standard proteinase K digestion and phenol/chloroform extraction. Then, the 150–200 bp DNA fragments library was analyzed with SureSelectXT HS kit and 67.3 Mb CRE v2 probes (Agilent). DNA fragments were barcoded, captured, and enriched by polymerase chain reaction (PCR) with SureSelect Post-Capture Primer Mix Agilent and P5 & P7 primers Illumina (P5: 5'-AATGATACGGCGACCACCGA-3'; P7: 5'-CAAGCAGAAGACGGCATACGAGAT-3'), before quantifying and qualifying using the TapeStation system. PCR was performed with an initial denaturation at 98°C for 2 min, followed by 9 cycles of 98°C for 30 s, 60°C for 30 s, and 72°C for 1 min, and a final extension at 72°C for 10 min. Fragments were then sequenced by NextSeq500 paired-end 2 × 100 bp (Illumina). Mean sequencing depths of 200× for somatic and 100× for germline were reached.

For the remaining 15 tumor samples and 2 tumor/normal pairs, sequence capture and exome sequencing were performed by IntegraGen SA (Evry, France). DNA was extracted from the FFPE tissue sample using the QIAamp Diagnostic Sample Preparation (DSP) DNA FFPE Tissue Kit (Qiagen, Hilden, Germany), according to the manufacturer's instructions. The protocol followed by IntegraGen has been described elsewhere [3]. In brief, genomic DNA was captured using Twist Human Core Exome Enrichment System (Twist Bioscience, South San Francisco, CA, USA)

+ IntegraGen Custom V2. Library Preparation Enzymatic Fragmentation Kit was used for library preparation. The eluted fraction was amplified by PCR with P5 & P7 primers TruSeq Illumina (Adapter P5: 5'-AATGATACGGCGACCACCGAGATCTACAC [INDEX] AACTCTTTCCCTACACGACGCTCTTCCGATCT-3' ---Insert; Adapter P7: 5'-CAAGCAGAAGACGGCATACGAGAT [INDEX] GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT-3' ---Insert). PCR amplification was carried out using the KAPA HiFi HotStart ReadyMix (2X) under the following conditions: an initial denaturation at 98°C for 45 s, followed by 7 cycles of 98°C for 15 s, 60°C for 30 s, and 72°C for 30 s, with a final extension at 72°C for 1 min. Fragments were then sequenced on an Illumina NovaSeq as Paired End 100 reads.

Bioinformatics and data analysis

Sample reads were aligned following Genome Analysis Toolkit (GATK) best practice with bwa version 0.7.17-r1188. Targeted regions were sequenced to an average depth of 150×, with 99% of the regions covered by $\geq 1\times$, 98% covered by $\geq 4\times$, and 95% covered by $\geq 20\times$.

For the detection of somatic single-nucleotide variants (SNVs) and base insertions or deletions (indels), we used GATK Mutect2. We applied the following filtering criteria for somatic mutation calling: alternative read count < 9. The following filtering criteria were applied for germline variant calling: alternative read count < 1. The variants were annotated using VEP (available at <http://www.ensembl.org/vep>) and SnpEff (available at <http://pcingola.github.io/SnpEff/>). Only frame-shift deletions/insertions, in-frame insertions/deletions, missense mutations, nonsense mutations and splice-site variants were selected.

For copy-number variation (CNV) analysis, we used CNVkit version 0.9.9 (available at <https://github.com/etal/cnvkit>). A homozygous deletion was considered when estimated log2 copy ratio was below 1. Intermediate estimated log2 copy ratios between -1 and -0.25 were considered as heterozygous deletions.

Tumor mutational burden (TMB) per Mb was assessed in samples with matched normal samples using the maftools R package version 2.10.05 and only non-synonymous variants with functional impact.

DNA methylation

Sample preparation and sequencing

Thirty-seven DNA samples were hybridized on the Infinium MethylationEPIC array (Illumina) according to the manufacturer's instructions at the Genomics and Proteomics Core Facility of the German Cancer Research Center (DKFZ) Heidelberg. Both fresh frozen and FFPE samples with tumor purity > 50% were included. DNA from frozen samples was extracted by standard proteinase K digestion and phenol/chloroform. Following deparaffinization, DNA from FFPE samples was extracted with the Maxwell® 16FFPE Plus LEV DNA Kit (Promega, Fitchburg, Wisconsin, USA) and a Maxwell DNA extractor (Promega).

Bioinformatics and data analysis

CNV analysis was performed using the 'conumee' R package in Bioconductor (available at <http://bioconductor.org/packages/conumee/>) from .IDAT files. In brief, the sum of the raw fluorescence signals from probes targeting both the methylated and unmethylated alleles of a CpG site was calculated, then the intensity values were log₂-transformed and referenced to the diploid genome. This copy-number ratio was then plotted in a graph according to chromosomal location. Homozygous deletion was considered when log₂ ratios of targeting probes were below 1. Intermediate log₂ ratios between -1 and -0.25 were considered as heterozygous deletions. A low-level copy number gain was defined as a log₂ ratio > 0.25 and an amplification as log₂ ratio > 0.4. Evaluation of arm-level copy number alteration was based on visual assessment of DNA methylation CNV plots.

Fluorescence hybridization in situ (FISH)

Sections (4 µm thick) were deparaffinized in xylene and rehydrated through a graded series of ethanol solutions. For the detection of SMARCB1 deletion, Zytolight® SPEC SMARCB1/22q12 Dual Color Probe was used (ZytoVision GmbH, Bremerhaven, Germany). We followed the procedure as recommended by the manufacturers and slides were viewed under a fluorescent microscope equipped with multi-bandpass filters to visualize colors simultaneously.

Clinical targeted next-generation sequencing (NGS)

A custom NGS panel known as DRAGON (Detection of Relevant Alterations in Genes involved in Oncogenetics by NGS), commercially available as SureSelect CD Curie CGP by Agilent, was performed in three cases at Institut Curie, Unit of Genetics (Paris, France). Library preparation and bioinformatics methods have been described elsewhere [4].

Single nucleotide polymorphism (SNP) array genomic profiling

SNP array genomic profiling was performed on three samples. The procedure was performed as previously described [5].

Shallow whole genome sequencing

DNA was extracted from FFPE tissue samples using the QIAamp DSP DNA FFPE Tissue Kit (Qiagen), following the manufacturer's instructions. The whole genome library was prepared by the SureSelect XT HS kit (Agilent) on NextSeq (Illumina). To predict sample tumor fraction and copy number counts, ichorCNA analysis was applied (available at <https://github.com/broadinstitute/ichorCNA>). IchorCNA uses a probabilistic model, implemented as a hidden Markov model (HMM), to simultaneously segment the genome, predict large-scale copy number alterations, and estimate the tumor fraction of an ultra-low-pass whole genome sequencing sample.

Immunohistochemistry analysis

Immunohistochemical staining was performed on 3 μ m-thick sections from FFPE tumor samples for antibodies to the following proteins: switch/sucrose non-fermentable-related matrix associated actin dependent regulator of chromatin subfamily B member 1 (SMARCB1), CD3, CD8, CD68, CD163, CD21, CD23 and desmoglein 2 (DSG2). The antibodies and conditions used for immunohistochemistry are summarized in **Supplementary Table S10**.

The density of tumor-infiltrating immune cells was evaluated on whole-tissue sections with a semi-quantitative method, as follows: 0: 0%; 1: 1%-25%; 2: 26%-50%; 3: > 50%, in the core of the tumor or intratumoral area and in the invasive margin or peritumoral area. The density of tumor-infiltrating immune cells was subsequently divided into high or low based on the median score as a cut-off for each individual immune cell.

Evaluation of mature tertiary lymphoid structure (TLS) was based on hematoxylin-eosin-saffron (HES) staining, CD20 and CD23 immunohistochemistry in non-lymphoid organs. Mature TLS was defined as a nodular CD20⁺ B lymphoid cell aggregate (including at least 50 immune cells) with CD23⁺ follicular dendritic cells [6].

Visium Spatial transcriptomics assay

Sample and library preparation

Representative FFPE tumor blocks from four EpS patients with DV200 > 30% were selected for the Visium Spatial transcriptomics assay (10x Genomics, Pleasanton, CA, USA) to provide optimal coverage of the 6.5 × 6.5 mm capture area. Using the human whole transcriptome probe set, spatial transcriptomics was performed with 10x Genomics Visium technology according to the Visium Spatial Gene expression for FFPE samples protocol described previously [7]. Generated Visium libraries were pooled and loaded in a single small paired-end Illumina flow cell. Sequencing was performed on NovaSeq PE 50 at a sequencing depth of 50,000 read pairs per spot covered with tissue using 10x Genomics recommended run parameters.

Bioinformatics and data analysis

Space Ranger (10x Genomics) was used to perform the demultiplexing of Visium-prepared raw base call files generated by Illumina sequencers into FASTQ files using Space Ranger version 2.0.0 mkfastq. Next, Space Ranger count was used to perform tissue and fiducial alignment from the microscope slide image and barcode/ Unique Molecular Identifier (UMI) counting to generate feature-barcode matrices that were further used for downstream analyses.

Downstream analyses relied on the Seurat R package (version 4.3, available at <https://satijalab.org/seurat/>). We calculated the average expression levels of selected genes subtracted by the aggregated expression of control gene sets using the AddModuleScore function from Seurat.

Human tumor samples scRNA-Seq analysis

Tumor dissociation and scRNA-Seq

For the tissue dissociation of the eight fresh tumor samples collected from seven patients with EpS, we followed the protocol as described previously [7]. Single cells were loaded into the 10x Genomics cassette for a targeted cell recovery of 5,000 cells per sample following the recommendations from the manufacturer's protocol. Two samples were profiled using the 10x Genomics Chromium Single Cell 3' GE kit and six samples using the Chromium Next GEM Single Cell 5' GE kit v1.1. Cell encapsulation, reverse transcription and library generation were performed according to 10x Genomics standard protocols [8, 9]. Paired-end sequencing was performed on the Illumina NovaSeq 6000 sequencer for a targeted depth of 400 million reads per sample.

Bioinformatics and data analysis

Raw Binary Base Call (BCL) files were demultiplexed and converted to Fastq format using bcl2fastq (version 2.20.0.422 from Illumina). Read quality control was performed using fastQC (version 0.11.9) and assignment to the expected genome species was evaluated with fastq-screen (version 0.14.0). Reads were pseudo-mapped to the Ensembl reference transcriptome v99 corresponding to the homo sapiens GRCH38 build with kallisto (version 0.46.2, available at <https://github.com/pachterlab/kallisto>) using its 'bus' subcommand and parameters corresponding to the 10x Chromium 5' scRNA-Seq v2 chemistry. The index was made with the kb-python (version 0.24.4) wrapper of kallisto. Barcode correction using a whitelist provided by the manufacturer (10x Genomics) and gene-based reads quantification were performed with BUSTools (version 0.40.0, available at <https://github.com/BUSTools/bustools>).

ScRNA-Seq data were analyzed using the Seurat R package (version 4.0.4). Samples stemming from different patients were combined by merging the raw count matrices of individual Seurat objects. The SCTransform normalization method [10] was used to normalize, scale, and select 2,000 highly variable genes. Pearson residuals from this regression were used for dimension reduction by principal component analysis (PCA). The number of PCA dimensions to keep for further analysis was evaluated by assessing a range of reduced PCA spaces using 3 to 50 dimensions, with a step of 2. For each generated PCA space, Louvain clustering of cells was performed using a range of values for the resolution parameter from 0.1 to 1.2 with a step of 0.1. The

optimal space was manually evaluated as the combination of kept dimensions and clustering resolution resolving the best structure (clusters homogeneity and compactness) in a uniform manifold approximation and projection (UMAP). Additionally, we used the clustree method (version 0.4.3) to assess if the selected optimal space corresponded to a relatively stable position in the clustering results tested for these dimensions/resolution combinations.

The integration of samples stemming from different patients was performed by using the Harmony algorithm (<https://github.com/immunogenomics/harmony>) [11]. Datasets were merged using the merge() function from Seurat (version 4.0.4), and the SCTransform normalization method was used to normalize, scale and select 2,000 highly variable genes. The reduced PCA spaces were used as input for the HarmonyMatrix() function implemented in Harmony package (version 0.1.0) where the batch effect was regressed. The batch-corrected shared space output by Harmony was used for clustering.

We used CerebroApp (version 1.3.1, available at <https://github.com/romanhaa/cerebroApp/>), a cell report browser which allows users to interactively visualize various parts of single-cell transcriptomics analyses to identify the most expressed genes in each cell subpopulation to compute pathway enrichment on marker genes (based on the Enrichr application programming interface) and Gene Set Enrichment Analysis (GSEA) [12].

Inference of single cells' CNV was performed using the InferCNV tool. Raw gene expression data were first extracted from each patient's Seurat object. For each sample, normal reference cells were selected based on the expression of immune cells and/or cancer-associated fibroblasts canonical markers and tumor cells were grouped according to annotated Louvain clusters as defined above. The cutoff for the minimum average read count per gene was set to 0.1, as recommended for 10x Genomics' data. The 'cluster_by_groups' setting was used to perform separate clustering for each cluster as defined in the cell annotations file. All other options were set to their default values. Each CNV was annotated as a gain or a loss to a p- or q-arm using the GRCh38 reference genome. We also applied InferCNV on the merged dataset with the same parameters, a down-sampling of 800 cells per cell type and using all the non-tumor cells as reference.

MoMac-VERSE atlas [13] was used to deconvolute and annotate myeloid cell subpopulations from the ScRNA-Seq individual dataset.

To infer cell-cell communication between identified tumor and immune cell clusters, we used CellphoneDB (available at <https://github.com/Teichlab/cellphonedb>), a repository of ligands, receptors and their interactions [14]. For this purpose, we integrated with Harmony, as previously described, the datasets from patients according to their molecular subtype. Cellphone DB was then applied separately in each EpS transcriptomic subtype integrated dataset. Mean is calculated as the average of the expression of the ligand in the source cell type and the receptor in the target cell type. The interactions with P -value <0.05 are considered significant.

Assessment of scRNA-Seq-derived signatures' specificity and prognostic value

CIBERSORTx

To deconvolve scRNA-Seq EpS malignant cell subclusters identified in EpS scRNA-seq data from EpS bulk RNA-seq, CIBERSORTx (<https://cibersortx.stanford.edu>) was used to derive a signature matrix from scRNA-seq data. Using the Seurat subset function, 500 or 1,000 cells were extracted from each cluster from the scRNA-seq integrated dataset. Cluster-labeled cells served to obtain a single-cell reference matrix (scREF-matrix) used as an input on the CIBERSORTx online server using the “Custom” option. Default values for replicates ($n = 5$), sampling (0.5), and minimum expression (0.0) were used. Additional options for kappa (999), q-value (0.01), and number of barcode genes (300–500) were kept at default values. The imputation of cell subtype fractions defined on the scREF-matrix was performed on counts per million (CPM) values from EpS bulk RNA-seq data using the ‘Impute Cell Fractions’ function with the ‘Custom’ option and ran in absolute mode. A total of 500 permutations was performed to test for statistical significance.

Gene signature scores calculation for specificity and prognostic assessment

The top 100 upregulated genes within each malignant cell subcluster from the scRNA-seq integrated dataset, defined by Seurat graph-based clustering using the K-nearest neighbor method, were selected to define cluster-specific gene signatures. A score for each cluster-specific signature was then calculated from the Institut Gustave Roussy EpS bulk RNA-seq dataset and an external cohort of 1,041 samples from 32

soft tissue tumor subtypes using the arithmetic mean of Z-scores of each gene expression expressed in variance stabilizing transformed raw counts. Samples with a number of unique reads less than 10 million were discarded from this analysis.

First, to assess the specificity of scRNA-Seq derived cluster-specific gene signatures for EpS, an analysis of variance (ANOVA) test was performed to compare the signature values between EpS transcriptomic subtypes, EERT and all other subtypes of soft tissue tumors. The significance test was set at $P < 0.05$.

To further evaluate the prognostic value of scRNA-Seq-derived cluster-specific gene signatures, the IGR EpS bulk RNA-seq dataset was segregated into ‘High’ and ‘Low’ signature scores using the optimal cutpoint method. We then performed a survival analysis for patients from whom the tumor samples were collected to compare ‘High’ versus ‘Low’ signature scores using the Kaplan-Meier model and tested for significance with the log-rank test. Overall survival was defined as the time from the EpS diagnosis until death from disease. Metastasis-free survival was defined as the time from the EpS diagnosis until the occurrence of distant metastasis or death from disease. Subsequently, we validated our findings on an external validation cohort of RNA-Seq data from 12 EpS samples.

Statistical analyses

Unless otherwise specified in the specific methods section, statistical data analysis was performed using PRISM 9 (GraphPad Software Inc., CA, USA) on the raw data. Comparisons between two groups were performed using the Fisher’s exact test for categorical variables and the Mann–Whitney U test for continuous (non-categorical) variables. A P -value < 0.05 was considered statistically significant.

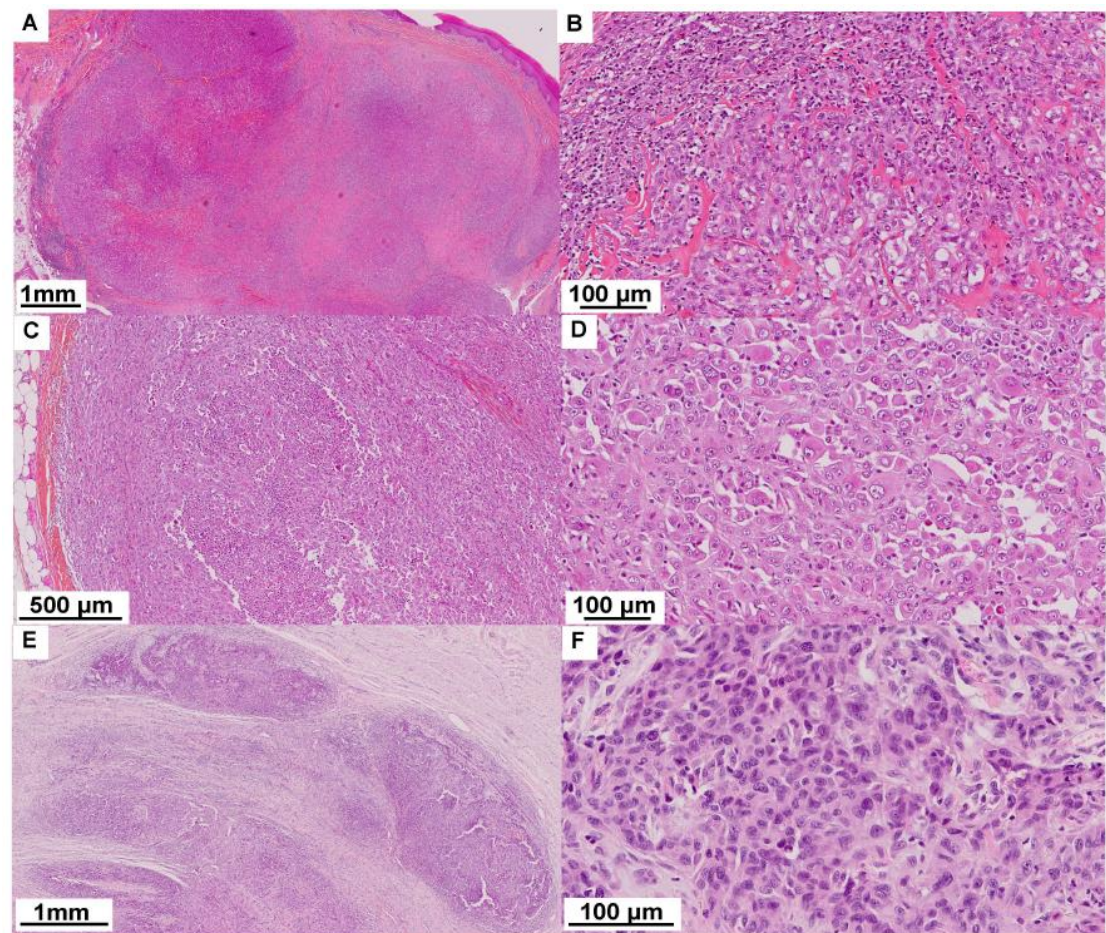
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Supplementary Figures

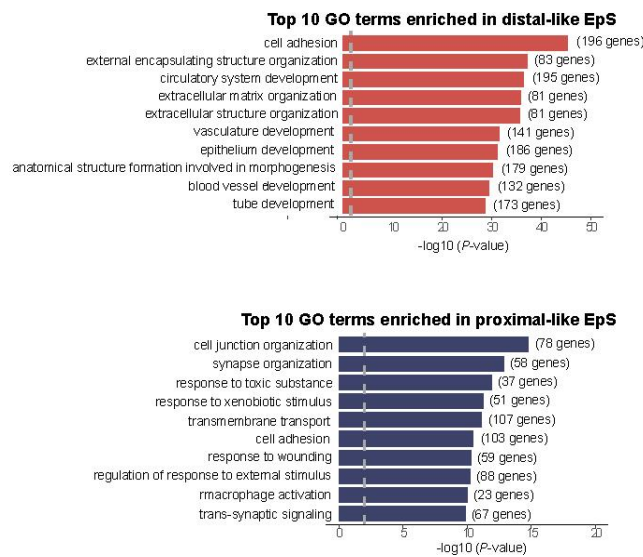


Supplementary Figure S1. Representative images of EpS histological subtypes

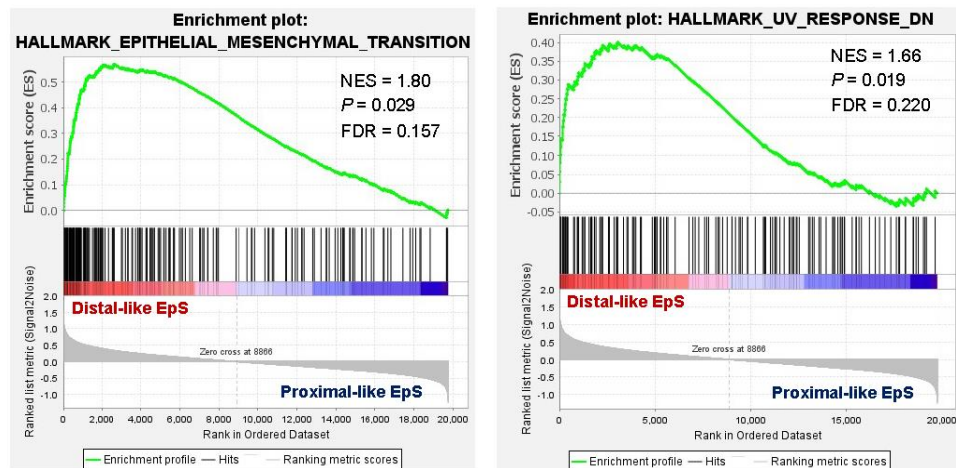
with HES staining. (A) The distal subtype of EpS consists of a pseudogranulomatous growth pattern of tumor cells with central degeneration. (B) Tumor cells in the distal subtype of EpS are epithelioid or spindle-shaped with mild atypia. (C) The proximal subtype of EpS consists of dense proliferation of large cells. (D) Proximal subtype of EpS with pleomorphic epithelioid cells, enlarged vesicular nuclei and prominent nucleoli. (E) Hybrid subtype of EpS displays features from both distal and proximal subtypes. The distal feature consists of a pseudogranulomatous nodular growth pattern. (F) The proximal feature in the hybrid subtype of EpS consists of areas of dense proliferation of large pleomorphic epithelioid cells with prominent nucleoli.

Abbreviations: EpS, epithelioid sarcoma, HES, hematoxylin-eosin-saffron

(A)



(B)

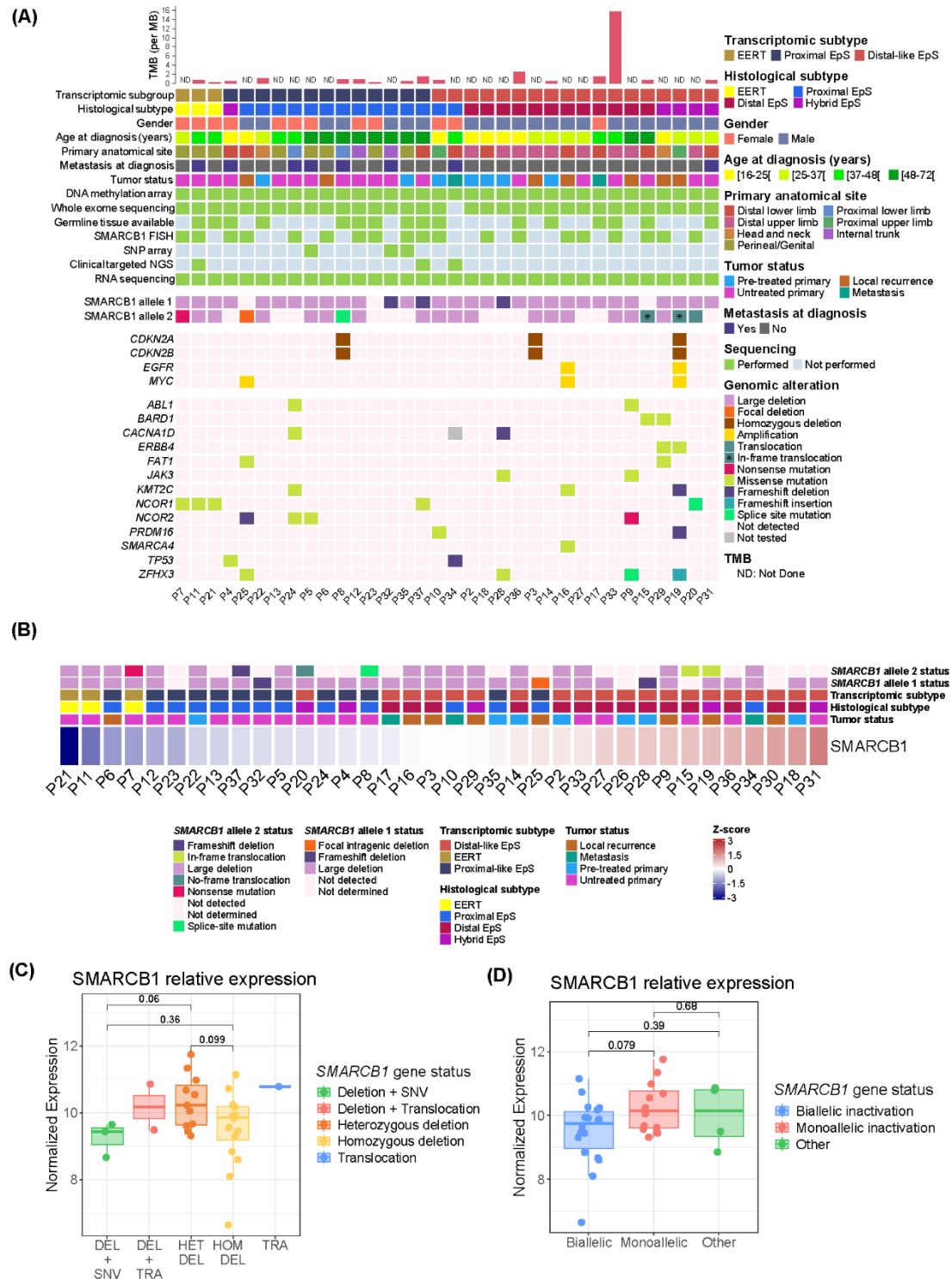


458

459 **Supplementary Figure S2. Transcriptome analysis identifies two molecular**
460 **subtypes of EpS with distinct biological features.** (A) Top 10 significantly enriched
461 GO terms related to biological processes between distal-like and proximal-like EpS
462 samples. The bar graph shows adjusted P -values and numbers of genes (in brackets)
463 mapped to GO terms (gray dashed line at $P = 0.05$). (B) Gene Set Enrichment
464 Analysis showing significant enrichment “Hallmark” gene sets related to epithelial-
465 mesenchymal transition and genes down-regulated in response to UV radiation in
466 distal-like EpS compared to proximal-like EpS. The significance criteria were
467 nominal P -value < 0.05 and FDR < 0.25 .

468 Abbreviations: EpS, epithelioid sarcoma; FDR, false discovery rate; GO, gene
469 ontology; NES, normalized enrichment score; UV, ultraviolet

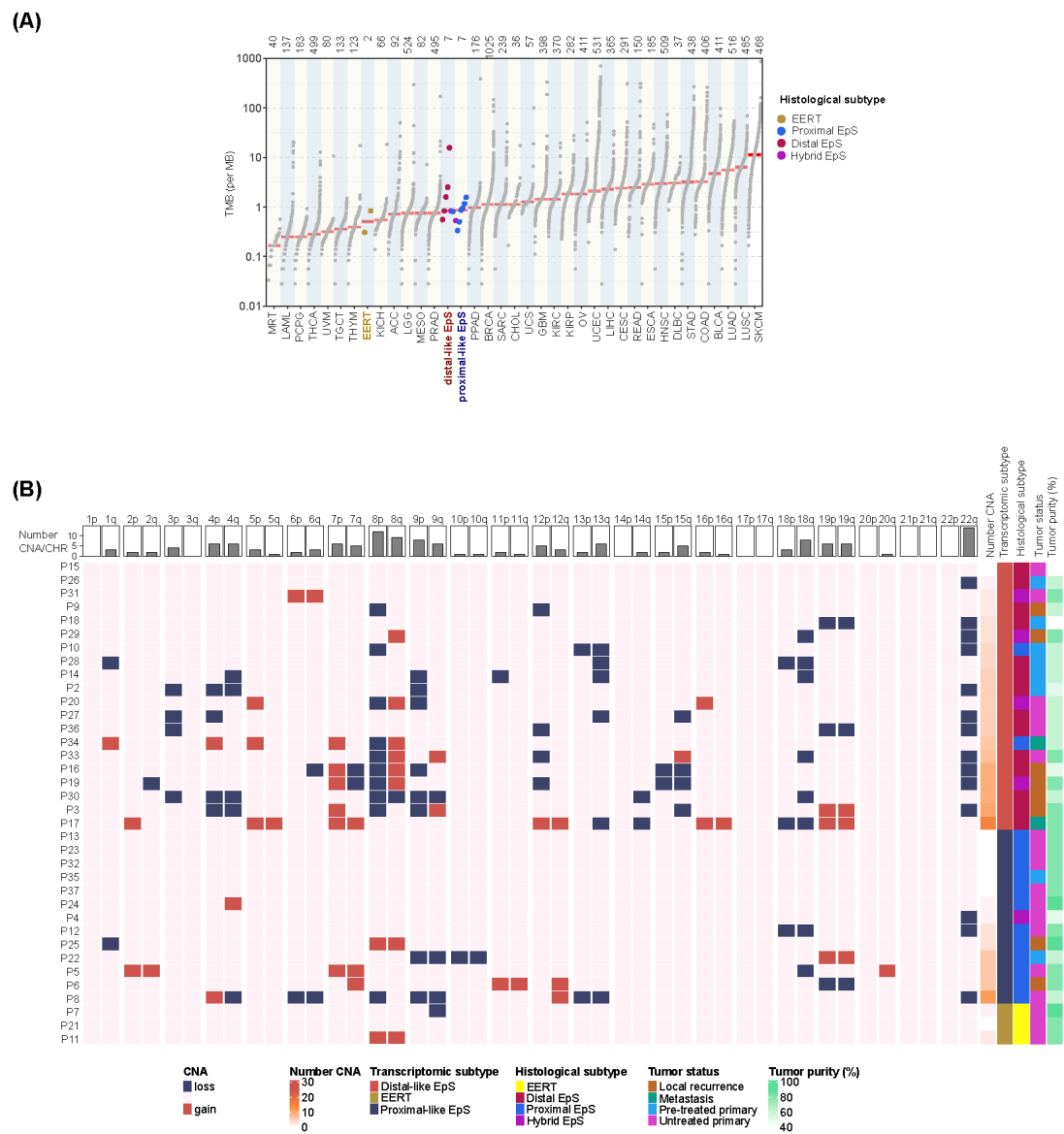
470



473 **Supplementary Figure S3. Somatic genomic alterations in EpS.** (A) Oncoplot
474 showing the clinical characteristics, assay used, *SMARCB1* gene status and selected
475 genomic alterations in distal-like EpS ($n = 18$), proximal-like EpS ($n = 13$) and EERT

($n = 3$). Each column represents a different patient and the sample ID is displayed at the bottom; only known tumor suppressors or oncogenes according to the catalogue of somatic mutations in cancer (COSMIC) database are shown. (B) *SMARCB1* mRNA levels across all samples, based on bulk RNA-Seq data. Heatmap shows Z-scores of *SMARCB1* gene expression across samples. (C) Boxplot displaying *SMARCB1* mRNA levels according to the type of *SMARCB1* alterations. The *P*-value was calculated using the Wilcoxon signed-rank test. (D) Boxplot displaying *SMARCB1* mRNA levels according to the type of *SMARCB1* gene alterations, comparing biallelic inactivation, monoallelic inactivation, and other *SMARCB1* alterations. The *P*-value was calculated using the Wilcoxon signed-rank test.

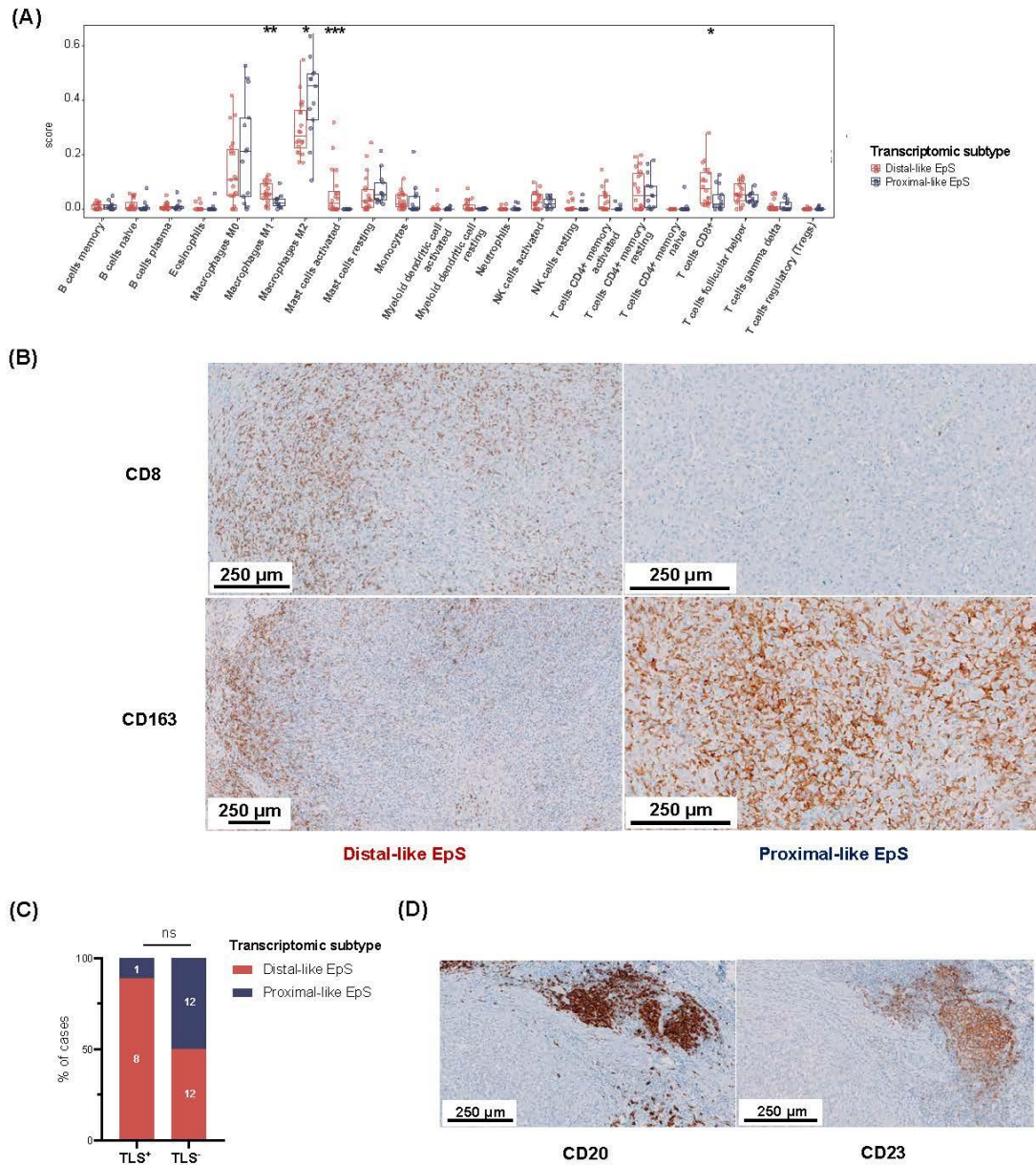
Abbreviations: BARD1, breast cancer type 1 associated RING domain 1; CACNA1D, calcium voltage-gated channel subunit alpha1; CDKN2A, cyclin-dependent kinase inhibitor 2A; CDKN2B, cyclin-dependent kinase inhibitor 2B; EERT: extracranial extrarenal rhabdoid tumor; EGFR, epidermal growth factor receptor; EpS, epithelioid sarcoma; ERBB4, v-erb-a erythroblastic leukemia viral oncogene homolog 4; FAT1, FAT atypical cadherin 1; FISH, fluorescence in situ hybridization; HET DEL, heterozygous deletion; HOM DEL, homozygous deletion; JAK3, Janus kinase 3; KAT2C, lysine acetyltransferase 2C; KMT2C, lysine methyltransferase 2C; NCOR1, nuclear receptor corepressor 1; NCOR2, nuclear receptor corepressor 2; PRDM16, positive regulatory domain I-binding factor 1 and retinoblastoma-interacting zinc finger protein 1/ su(var)3-9, enhancer-of-zeste, trithorax domain 16; SMARCA4, switch/sucrose non-fermentable-related, matrix associated, actin dependent regulator of chromatin, subfamily A, member 4; SMARCB1, switch/sucrose non-fermentable-related matrix associated actin dependent regulator of chromatin subfamily B member 1; SNP, single nucleotide polymorphism; SNV, single nucleotide variant; TP53, tumor protein p53; TRA, translocation; ZFHX3, zinc finger homeobox 3



Supplementary Figure S4. Mutational and copy-number landscape of the two molecular subtypes of EpS. (A) Scatter plot comparing the non-synonymous TMB per genome of distal-like EpS ($n = 6$), proximal-like EpS ($n = 6$) and EERT ($n = 2$) with other tumors. Malignant rhabdoid tumors' data are from Chun et al. [15] and other tumor types are based on the TCGA data (<http://cancergenome.nih.gov/>). Tumor types are ordered by their median mutation load, shown as a horizontal red line. (B) Arm-level gains and losses based on the DNA methylation data in distal-like EpS ($n = 20$), proximal-like EpS ($n = 13$) and EERT ($n = 3$). Blue corresponds to a loss of at least one copy and red corresponds to a gain.

Abbreviations: ACC, adrenocortical carcinoma; BLCA, bladder urothelial carcinoma; BRCA, breast invasive carcinoma; CESC, cervical squamous cell carcinoma &

516 endocervical adenocarcinoma; CHOL, cholangiocarcinoma; CHR, chromosome;
517 CNA, copy number alteration; COAD, colon adenocarcinoma; DLBC, lymphoid
518 neoplasm diffuse large B-cell lymphoma; EERT: extracranial extrarenal rhabdoid
519 tumor; EpS, epithelioid sarcoma; ERBB4, GBM, glioblastoma; HNSC, head & neck
520 squamous cell carcinoma; KICH, kidney chromophobe; KIRC, kidney renal clear cell
521 carcinoma; KIRP, kidney renal papillary cell carcinoma; LAML, acute myeloid
522 leukemia; LGG, low grade glioma; LIHC, hepatocellular carcinoma; LUAD, lung
523 adenocarcinoma; LUSC, lung squamous cell carcinoma; MESO, mesothelioma;
524 MRT, malignant rhabdoid tumor; OV, ovarian serous cystadenocarcinoma; PAAD,
525 pancreatic adenocarcinoma; PCPG, pheochromocytoma & paraganglioma; PRAD,
526 prostate adenocarcinoma; READ, rectal adenocarcinoma; SARC, sarcoma; SKCM,
527 skin cutaneous melanoma; STAD, stomach adenocarcinoma; TCGA, the cancer
528 genome atlas; TGCT, testicular germ cell tumors; THCA, papillary thyroid
529 carcinoma; THYM, thymoma; TMB, tumor mutational burden; UCEC, uterine corpus
530 endometrial carcinoma; UCS, uterine carcinosarcoma; UVM, uveal Melanoma
531

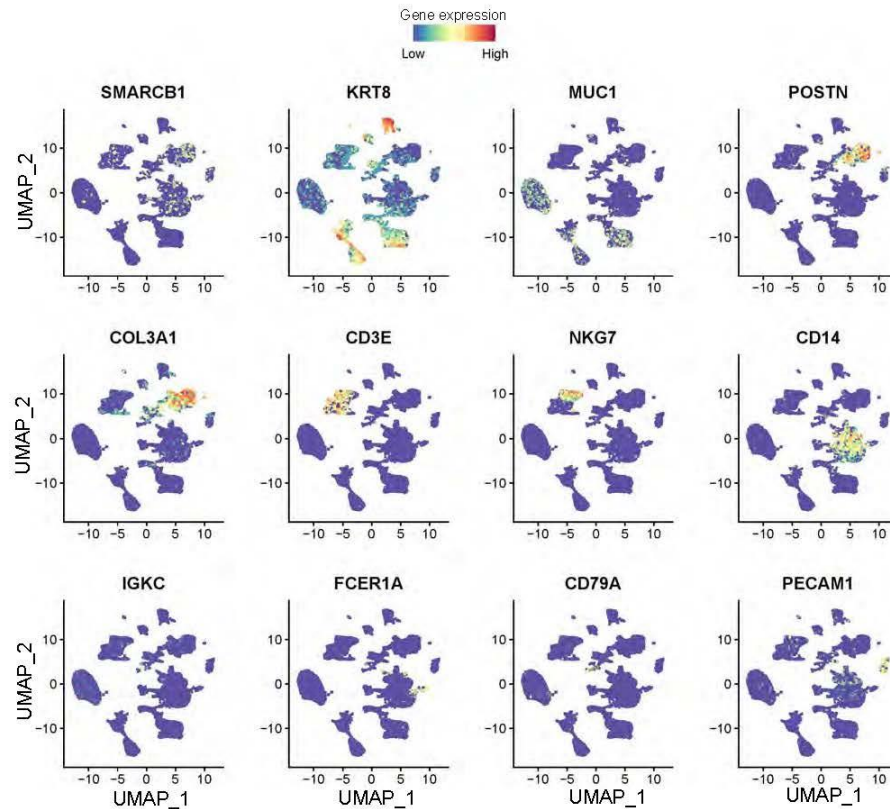


Supplementary Figure S5. Differences in immune cell composition between the two molecular subtypes of EpS. (A) Boxplots comparing the means of CIBERSORT-estimated abundances for 22 immune cell types between distal-like and proximal-like EpS subtypes in the bulk RNA-seq EpS cohort. The lower and upper hinges correspond to the first and third quartiles (the 25th and 75th percentiles). The upper whisker extends from the hinge to the largest value no further than $1.5 \times \text{IQR}$ from the hinge (where IQR is the inter-quartile range or distance between the first and third quartiles). The lower whisker extends from the hinge to the smallest value at most $1.5 \times \text{IQR}$ of the hinge. * is for $P < 0.05$, ** is for $P < 0.01$ and *** is for $P < 0.001$ (Wilcoxon signed-rank test). (B) Example of IHC staining for CD8 in T cells

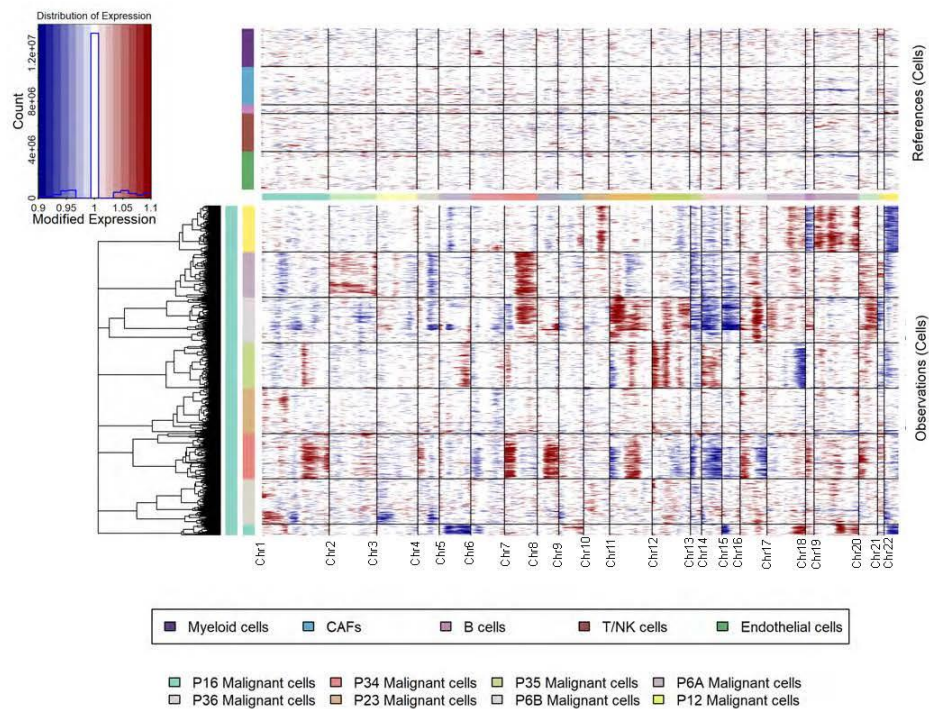
and CD163 in macrophages at the periphery of a nodule of tumor cells in a distal-like EpS sample (left panel) and in the tumor core of a proximal-like EpS sample (right panel). (C) Distribution of distal-like and proximal-like EpS transcriptomic subtypes in mature TLS⁺ and TLS⁻ groups. (D) Representative images of CD20 (left) and CD23 (right) immunostaining in a mature TLS in EpS.

Abbreviations: EpS: epithelioid sarcoma; IHC: immunohistochemistry; NK, natural killer; TLS: tertiary lymphoid structure

(A)



(B)



552

553 **Supplementary Figure S6. ScRNA-seq recapitulates EpS cellular composition.**

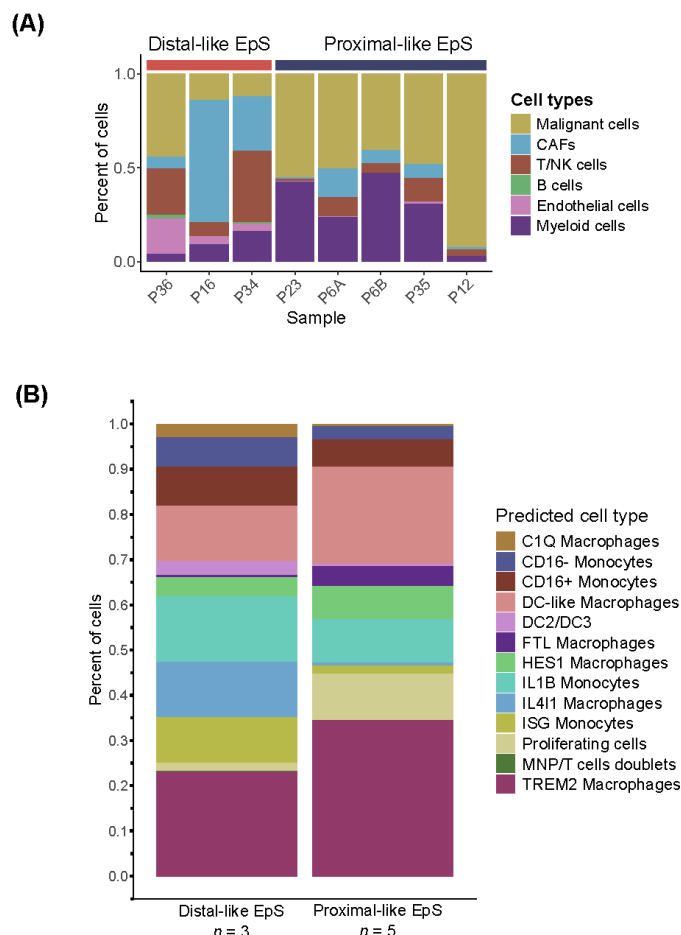
554 (A) UMAP showing single-cell profiles of malignant and non-malignant cells from 8
555 EpS colored by expression of selected genes (tumor cell markers: *SMARCB1*, *KRT8*,
556 *MUC1*; CAF markers: *POSTN*, *COL3A1*; NK/T cells markers: *CD3E*, *NKG7*;

557 myeloid cells markers: *CD14*, *FCER1A*; B cell markers: *IGKC*, *CD79A*; endothelial
558 cells marker: *PECAMI*). (B) Single-cell level copy number variation inference using
559 the InferCNV tool on the merge of individual scRNA-Seq datasets.

560 Abbreviations: CAF: cancer-associated fibroblast; COL3A1, collagen type III alpha 1
561 chain; EpS, epithelioid sarcoma; FCER1A, fc epsilon receptor Ia; IGKC,
562 immunoglobulin kappa constant; KRT8, keratin 8; MUC1, mucin 1; NK, natural
563 killer; NKG7, natural killer cell granule protein 7; PECAM1, platelet endothelial cell
564 adhesion molecule; POSTN, periostin; scRNA-Seq, single-cell RNA sequencing;
565 SMARCB1, switch/sucrose non-fermentable-related matrix associated actin
566 dependent regulator of chromatin subfamily B member 1; UMAP: uniform manifold
567 approximation and projection

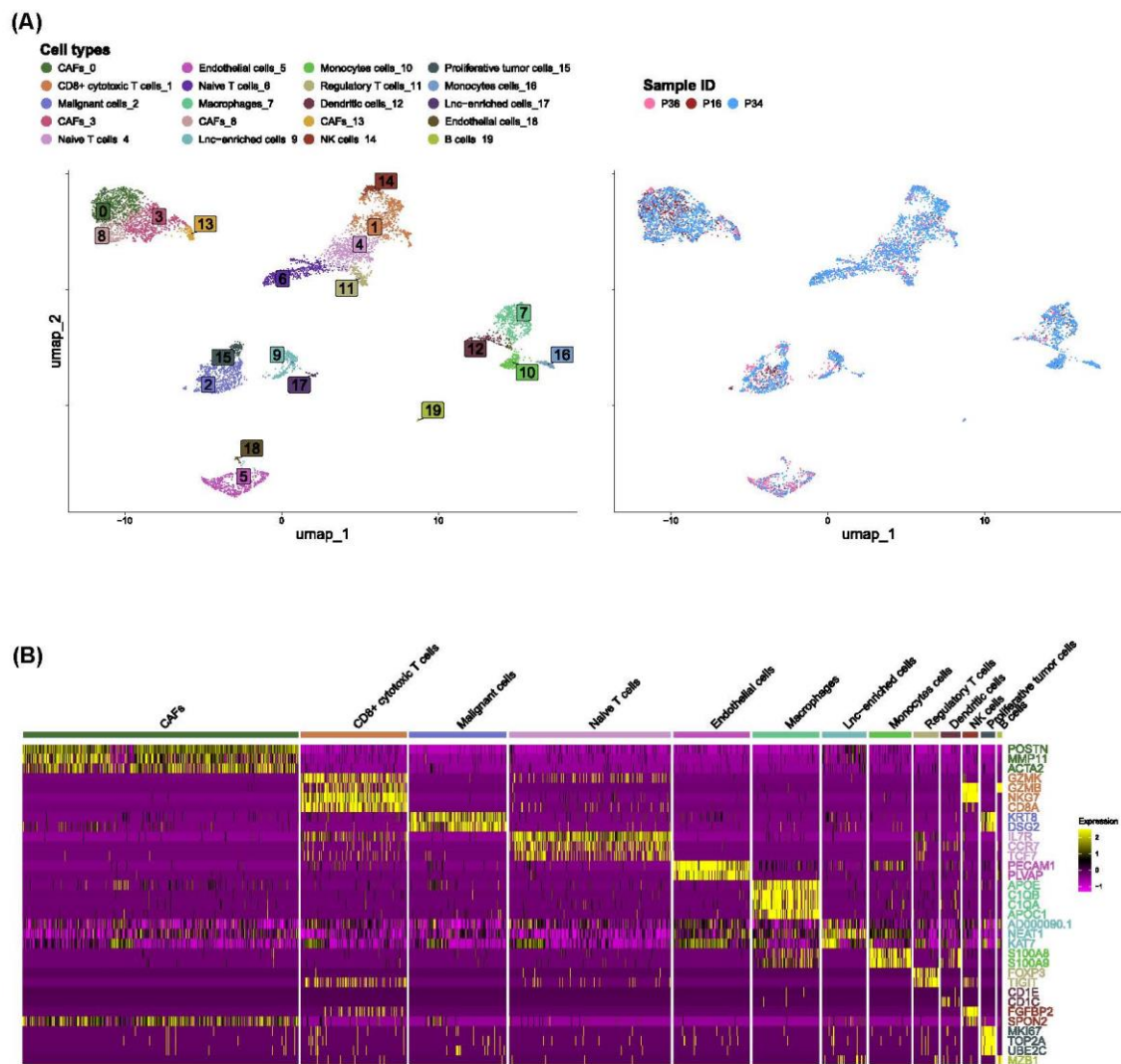
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569



Supplementary Figure S7. Myeloid cells are the most abundant immune subpopulation in EpS. (A) Barplot highlighting the proportion of each EpS cell subpopulation in the individual scRNA-Seq datasets. (B) Barplot highlighting the proportion of infiltrating myeloid cells in EpS from the scRNA-Seq individual datasets according to transcriptomic subtype. Myeloid cells are annotated according to the MoMac-VERSE atlas [13].

Abbreviations: CAF, cancer-associated fibroblast; DC, dendritic cell; EpS, epithelioid sarcoma; FTL, ferritin light chain; HES1, hairy and enhancer of split 1; IL1B, interleukin 1 beta; IL4I1, interleukin 4 induced 1; ISG, interferon-stimulated gene; MNP, mononuclear phagocyte; NK, natural killer; TREM2, triggering receptor expressed on myeloid cells 2



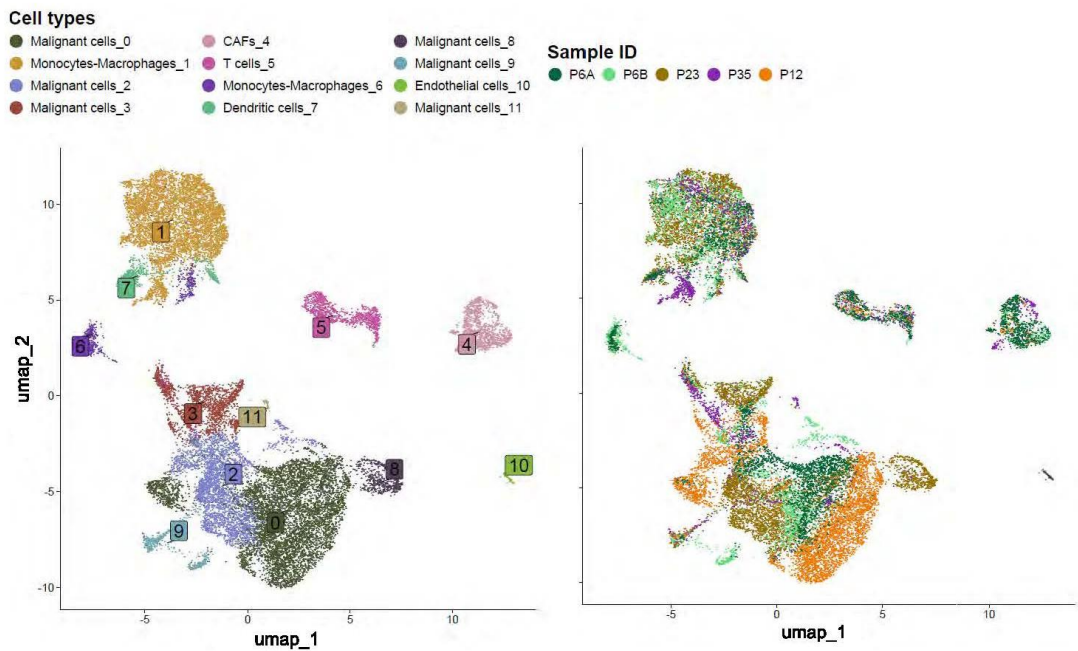
Supplementary Figure S8. ScRNA-Seq data related to ligand-receptor interactions between tumor and immune cells in distal-like EpS subtype. (A) UMAP showing integrated single-cell (Int-sc) profiles of distal-like EpS malignant and non-malignant cells colored by cell identity (left panel) and sample of origin (right panel). (B) Heatmap showing expression Z-score of selected marker genes in distal-like EpS Int-sc cell types.

Abbreviations: ACTA2, actin alpha 2; APOC1, apolipoprotein C1; APOE, apolipoprotein E; CAF: cancer-associated fibroblast; CCR7, C-C chemokine receptor type 7; C1QA, complement C1q A Chain; C1QB, complement C1q B Chain; DSG2, desmoglein 2; EpS, epithelioid sarcoma; FGFBP2, fibroblast growth factor binding protein 2; FOXP3, forkhead box P3; GZMB, granzyme B; GZMK, granzyme K; IL7R, interleukin 7 receptor; Int-sc, integrated single-cell; KAT7, lysine acetyltransferase 7; KRT8, keratin 8; Lnc: long non-coding; MKI67, marker of

597 proliferation Ki-67; MMP11, matrix metalloproteinase 11; MZB1, marginal zone B
598 and B1 cell-specific protein 1; NEAT1, nuclear paraspeckle assembly transcript 1;
599 NK, natural killer; NKG7, natural killer cell granule protein 7; PECAM1, platelet
600 endothelial cell adhesion molecule; PLVAP, plasmalemma vesicle associated protein;
601 POSTN, periostin; S100A8, S100 calcium-binding protein A8; S100A9, S100
602 calcium-binding protein A9; ScRNA-Seq, single-cell RNA sequencing; SPON2,
603 spondin 2; TCF7, transcription factor 7; TGIT, T-cell immunoreceptor with
604 immunoglobulin and immunoreceptor tyrosine-based inhibitory motif domains;
605 TOP2A, DNA topoisomerase 2-alpha; UBE2C, Ubiquitin Conjugating Enzyme E2 C;
606 UMAP: uniform manifold approximation and projection;

607

(A)



(B)

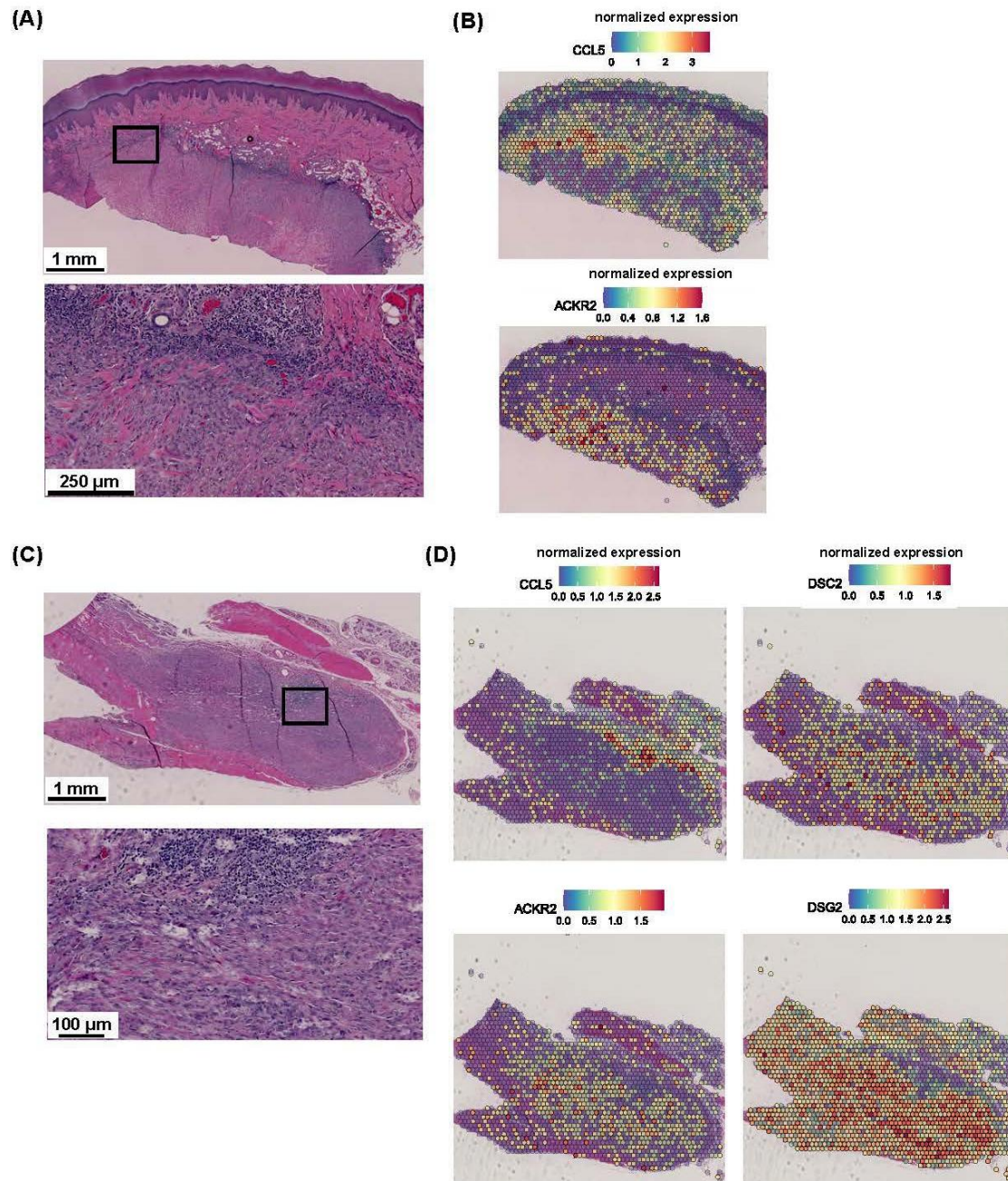


608

609 **Supplementary Figure S9. ScRNA-Seq data related to ligand-receptor**
610 **interactions between tumor and immune cells in proximal-like EpS molecular**
611 **subtype. (A) UMAP showing Int-sc profiles of proximal-like EpS malignant and non-**
612 **malignant cells colored by cell identity (left panel) and sample of origin (right panel).**
613 **(B) Heatmap showing expression Z-score of selected marker genes in proximal-like**
614 **EpS Int-sc cell types.**

615 Abbreviations: APOC1, apolipoprotein C1; APOE, apolipoprotein E; C1QA,
616 complement C1q A Chain; C1QB, complement C1q B Chain; CAF: cancer-associated
617 fibroblast; EpS, epithelioid sarcoma; GZMA, granzyme 1; Int-sc, integrated single-
618 cell; KRT8, keratin 8; KRT19, keratin 19; MMP11, matrix metalloproteinase 11;
619 NKG7, natural killer cell granule protein 7; PECAM1, platelet endothelial cell
620 adhesion molecule; PLVAP, plasmalemma vesicle associated protein; POSTN,
621 periostin; UMAP: uniform manifold approximation and projection

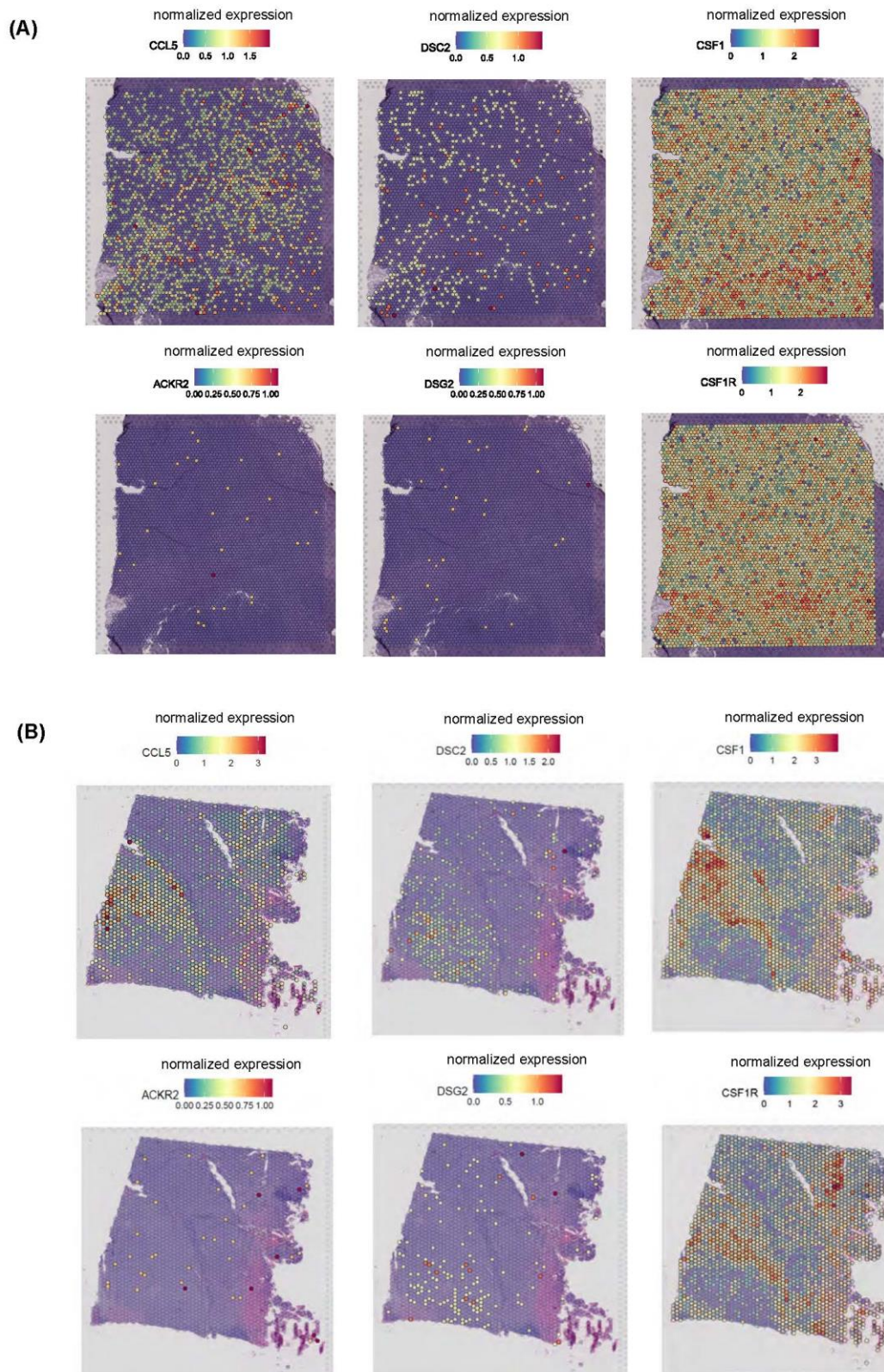
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Supplementary Figure S10. Spatial distribution of selected ligand-receptor pairs involved in the intercellular interactions between the immune and tumor cells, in distal-like EpS using Visium Spatial transcriptomics (10x Genomics). (A) HES-stained slide of the capture area by the Visium assay, represented in Figure 1G, from P36's distal-like EpS sample. The lower panel corresponds to a higher magnification of the boxed area. Immune cell infiltration was seen at the periphery of the tumor. (B) Spatial representation of the ligand CCL5 and its receptor ACKR2 expression levels by Visium assay in the P36 distal-like EpS sample. (C) HES-stained slide of the

capture area by Visium assay from P33 distal-like EpS sample. The lower panel corresponds to a higher magnification of the boxed area. Immune cell infiltration was seen at the periphery of the tumor. (D) Spatial representation of the ligands CCL5 and DSC2 and their cognate receptors ACKR2 and DSG2 expression levels by Visium assay in the P33 distal-like EpS sample.

Abbreviations: ACKR2, atypical chemokine receptor 2; CCL5, C-C motif chemokine ligand 5; DSC2, desmocollin 2; DSG2, desmoglein 2; EpS, epithelioid sarcoma; HES, hematoxylin-eosin-saffron

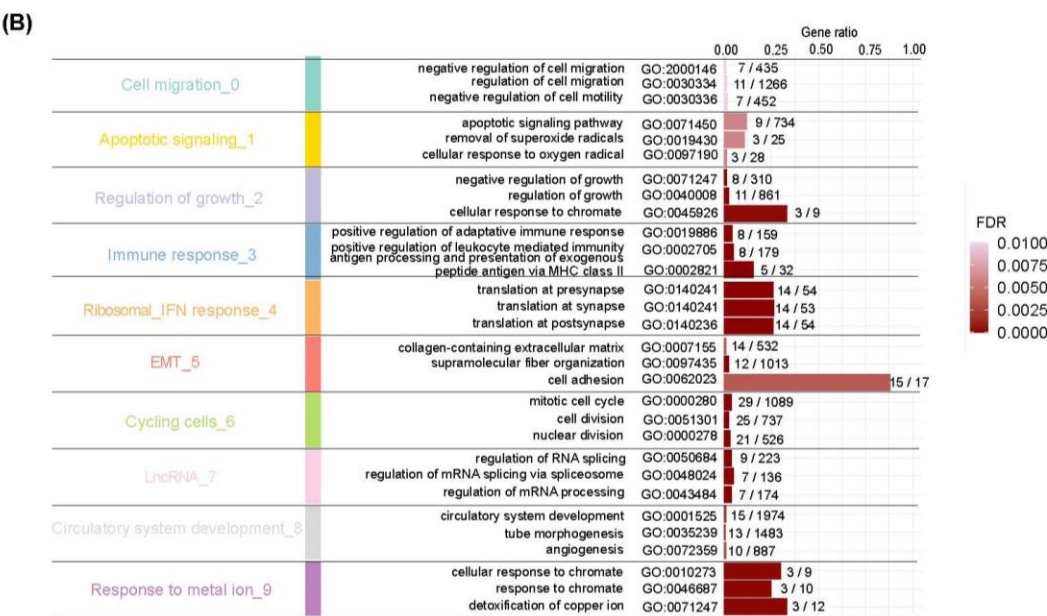
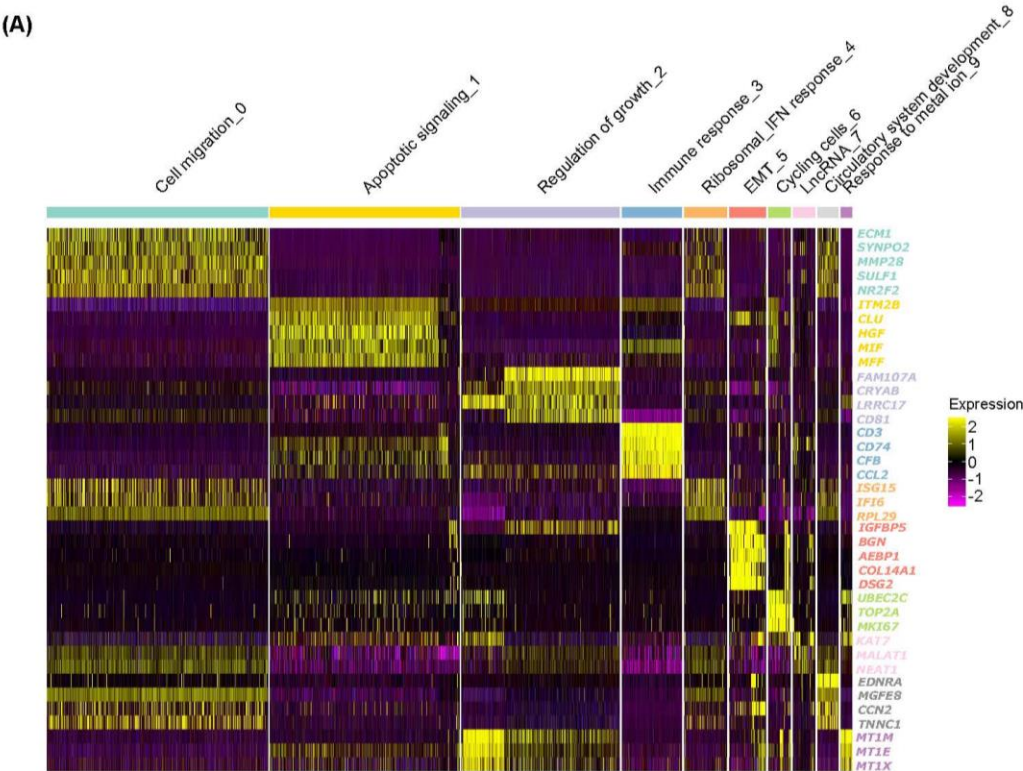


641

642 **Supplementary Figure S11. Spatial distribution of selected ligand-receptor pairs**
 643 **involved in the intercellular interactions between the immune and tumor cells, in**

644 **proximal-like EpS using Visium Spatial transcriptomics (10x Genomics).** (A)
645 Spatial representation of the ligands CCL5, DSC2 and CSF1 and their cognate
646 receptors ACKR2, DSG2 and CSF1R expression levels by Visium assay in the P35
647 proximal-like EpS sample. (B) Spatial representation of the ligands CCL5, DSC2 and
648 CSF1 and their cognate receptors ACKR2, DSG2 and CSF1R expression levels by
649 Visium assay in the P23 proximal-like EpS sample.

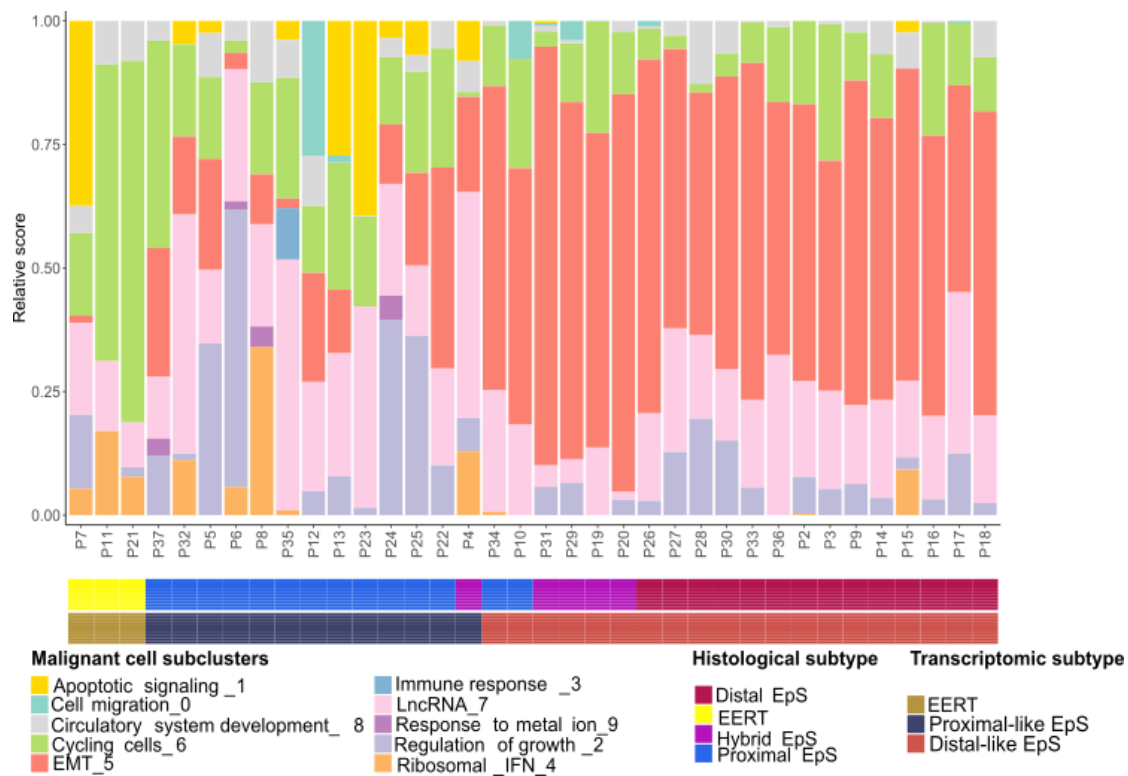
650 Abbreviations: ACKR2, atypical chemokine receptor 2; CCL5, C-C motif chemokine
651 ligand 5; CSF1, colony-stimulating factor 1; CSF1R, colony stimulating factor 1
652 receptor; DSC2, desmocollin 2; DSG2, desmoglein 2; EpS, epithelioid sarcoma
653



Supplementary Figure S12. ScRNA-Seq highlights a cancer cell-specific EMT signature of distal-like EpS molecular subtype. (A) Heatmap showing expression Z-score of the top 50 differentially expressed genes of each EpS malignant cell subcluster. Key markers of each subcluster are shown on the right. (B) Top three significantly enriched GO pathways in each single-cell EpS malignant cell subcluster.

660 Abbreviations: AEBP1, adipocyte enhancer-binding protein 1; BGN, biglycan; CCL2,
 661 C-C motif chemokine ligand 2; CCN2, cellular communication network factor 2;
 662 CFB, complement factor B; CLU, clusterin; COL14A1, collagen type XIV alpha 1
 663 chain; CRYAB, crystallin alpha B; DSG2, desmoglein 2; ECM1, extracellular matrix
 664 protein 1; EDNRA, endothelin receptor type A; EMT: epithelial- mesenchymal
 665 transition; FAM107A, family with sequence similarity 107 member A; FDR, false
 666 discovery rate; HGF, hepatocyte growth factor ; IFI6, interferon alpha inducible
 667 protein 6; IFN: interferon; ISG15, interferon-stimulated gene 15; ITM2B, integral
 668 membrane protein 2B; KAT7, lysine acetyltransferase 7; LncRNA: long non-coding
 669 RNA LRRC17, leucine Rich Repeat Containing 17; MALAT1, metastasis associated
 670 lung adenocarcinoma transcript 1; MFF, mitochondrial fission factor; MGFE8, milk
 671 fat globule-EGF and factor V/VIII domain containing; MIF, macrophage migration
 672 factor; MKI67, marker of proliferation Ki-67; MMP28, matrix metalloproteinase 28;
 673 MT1E, metallothionein 1E, MT1M, metallothionein 1M; MT1X, metallothionein 1X;
 674 NEAT1, nuclear paraspeckle assembly transcript 1; NR2F2, nuclear receptor
 675 subfamily 2 group F member 2; RPL29, ribosomal protein L29; SULF1, sulfatase 1;
 676 SYNPO2, synaptopodin 2; TOP2A, DNA topoisomerase II Alpha; TNNC1, troponin
 677 C1; UBEC2C, Ubiquitin Conjugating Enzyme E2 C; UMAP: uniform manifold
 678 approximation and projection

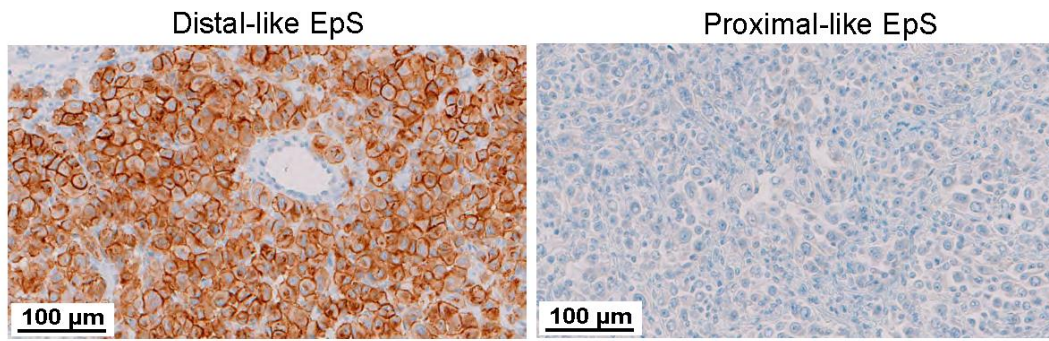
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Supplementary Figure S13. CIBERSORTx deconvolution of scRNA-Seq EpS malignant cell subclusters from the bulk RNA-seq EpS cohort.

Abbreviations: EpS, epithelioid sarcoma; EMT: epithelial-mesenchymal transition; IFN: interferon; LncRNA: long non-coding RNA

DSG2 IHC



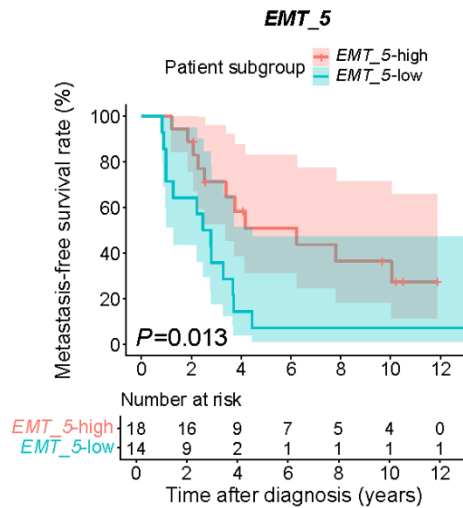
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686 **Supplementary Figure S14. DSG2 immunostaining.** Representative example of a
687 positive IHC staining for DSG2 in a distal-like EpS sample (left panel) and a negative
688 IHC staining for DSG2 in a proximal-like EpS sample (right panel).

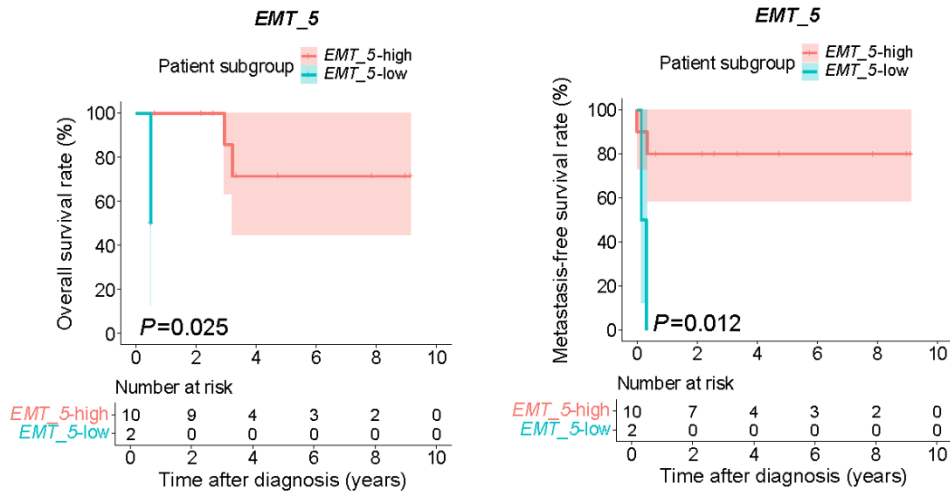
689 Abbreviations: DSG2, desmoglein 2; EpS, epithelioid sarcoma; IHC,
690 immunohistochemistry

691

(A)



(B)



692

693 **Supplementary Figure S15. Assessment of single-cell derived *EMT_5* signature's**
694 **prognostic value.** (A) Kaplan-Meier metastasis-free survival analysis according to
695 high and low *EMT_5* signature score in the bulk RNA-Seq dataset of the EpS cohort.
696 B) Kaplan-Meier overall survival (left panel) and metastasis-free survival (right
697 panel) analyses according to high and low *EMT_5* signature score in an independent
698 external RNA-Seq cohort of 12 EpS. *P* values are calculated using the Log-rank test.

699 Abbreviation: EMT: epithelial-mesenchymal transition